

CS 6603: AI, Ethics, and Society
Homework Project #2: Stats 101 Assignment

Readings:

- Chapter 4: Weapons of Math Destruction (Propaganda Machine: Online Advertising)
- *Optional:* Darrel Huff, Chapter 1: How to lie with statistics, Norton, New York, 1954 - [Internet Search – Huff + “How to lie with statistics” + pdf]

In this assignment, you’ll begin the process of exploring relationships in data. You’ll accomplish this task by computing some basic statistical measures on one of three datasets. This is a good time to learn or reboot your Python coding skills.

As you complete the steps below, please make sure that the *report outputs* correspond to the sections in your Assignment. (ex. Report output 1.1, should match to section 1.1 in your assignment).

Step 1: Select the dataset

Review the three datasets available on CANVAS. Select one that you will complete the assignment with.

- [mental-health-in-tech-survey-2019.csv] Mental Health in Tech Survey: Survey on Mental Health in the Tech Workplace in 2019 - <https://osmhhelp.org/research.html>
Dependent Variables:
 - *treatment*: Have you ever sought treatment for a mental health disorder from a mental health professional? (FALSE/TRUE)
 - *mental_health_disclosure*: Would you feel comfortable discussing a mental health issue with your direct supervisor(s)? (Yes/Maybe/No)
 - *mental_health_support*: Overall, how well do you think the tech industry supports employees with mental health issues? (numerical: 1-5)
- [hospital-discharge-2017.csv] New York Hospital Inpatient Discharge Information in 2017: <https://health.data.ny.gov/dataset/Hospital-Inpatient-Discharges-SPARCS-De-Identified/22g3-z7e7>
Dependent Variables:
 - *Length of Stay*: a numeric value representing number of days between admission and discharge
 - *APR Severity of Illness Code*: a numeric value representing severity of illness
- [deaths-in-custody.csv] Information on deaths that occur in custody or during the process of arrest in California - <https://openjustice.doj.ca.gov/data>
Dependent Variables:
 - *manner_of_death*: Suicide/Natural/Accidental/Homicide/Cannot be Determined/Other (for all others)
 - *custody_status*: Sentenced/Awaiting Booking/Booked – Awaiting Trial/ Booked – No Charges Filed/Other (for all others)

Report output: In your report, please document:

1.1. Dataset selected

Step 2: Explore the dataset

Review the dataset you have selected. Determine whether any data cleanup is needed. If so, perform the necessary cleanup steps, then answer the following questions.

Report output: In your report, please answer:

- 2.1. Did you perform data cleanup for this dataset?
If yes, outline the cleanup procedures you applied and report any resulting modifications to the data (e.g., changes in the number of records/observations or variables). If no, provide your reasoning for not performing cleanup.
- 2.2. How many observations are in the dataset?
If cleanup was performed, report the count after cleaning; otherwise, report the original total.
- 2.3. How many variables are in the dataset?
If cleanup was performed, report the count after cleaning; otherwise, report the original total.
- 2.4. Does this dataset seem to belong to a regulated domain in law (as discussed in the lectures)?
Explain why or why not.
- 2.5. In a table format, list any variable associated with a legally protected class. For each variable, identify the protected class and the associated legal precedence/law.

Example Output (Note: This is a partial example that uses a different dataset)

Data cleaned: Yes, I performed ...

Number of Observations: 1,400

Number of Variables: 16

Regulated Domain in Law: Housing (Fair Housing Act), because...

Protected Class Table

Variable	Protected Class	Law
nationality	National origin	Civil Rights Act of 1964, 1991
pregnant (y/n)	Pregnancy	Pregnancy Discrimination Act

Step 3: Determine the relationships between dependent and independent variables

The frequency of a value represents the number of times a value occurs in a data set. Compute the frequency of each value associated with each dependent variable (listed in Step 1) as a function of all of the protected class variables (independent variables) identified in Step 2. Create table(s) and histogram(s) comparing the frequency values of the dependent variable as a function of the independent variable.

Notes:

- For variables that are continuous, you might consider creating intervals that represent the data.
- For categorical/ordinal/nominal values, you might consider converting to numerical values based on a reasonable (albeit subjective) ordering.
- Step 3 should be performed for **each** dependent variable against all of the protected class variables. For example, if the dataset has 3 protected class variables and 2 dependent variables, there should be 6 sets of histograms **and** frequency tables.
- If you identified more than 3 protected class variables, select only 3 protected class variables to compute for this step. Please specify which 3 protected class variables you have selected.

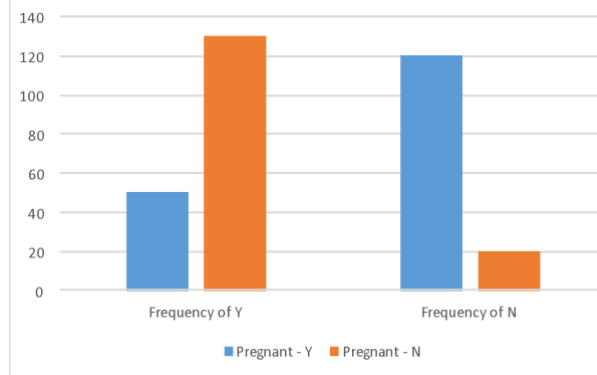
Report output: In your report, please document your findings:

- 3.1. The sets of frequency table(s) and histogram(s)

Example Output for One Dependent-Independent Variable Combination:

Pregnancy Status	Housing Decision (Y)	Housing Decision (N)	Total
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Pregnant – Y	50	120	170
Pregnant – N	130	20	150
Total	180	140	320



Step 4: Manipulation of data

In lecture, we learned how data can be manipulated in graphs to mislead users. For this step:

- Select one independent variable (protected class variable) and one dependent variable
- Create a graph to support a “fairness” hypothesis (e.g. the system is fair; there is no difference in the outcomes.)
- Create a graph to support a “bias” hypothesis (e.g. the system is biased; there is a difference in the outcomes.)

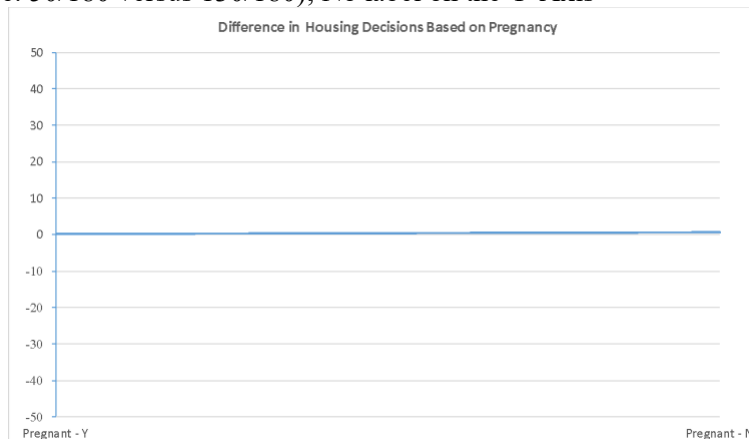
Report output:

- 4.1. For each, provide the hypothesis, the graph and a **detailed** description of any data manipulations performed to produce the graph and support the hypothesis.

Example Output:

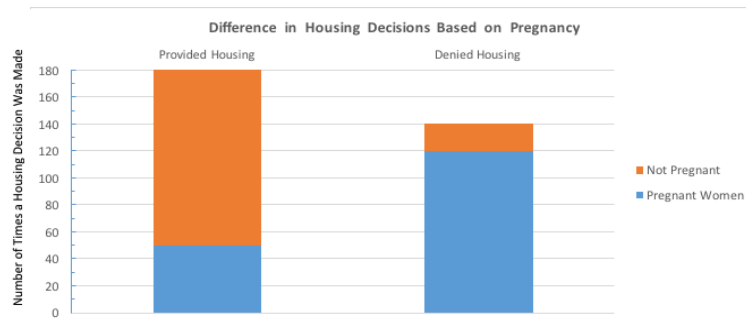
- 1) Fair Hypothesis: As seen from this graph, housing decisions are not dependent on the pregnancy status of women.

Manipulations: Used a line graph; Increased Scale to +-50; Mapped the ratio of positive Y decisions (i.e. 50/180 versus 130/180); No label on the Y-Axis



- 2) Bias Hypothesis: As seen from this graph, housing decisions are significantly dependent on the pregnancy status of women.

Manipulations: This hypothesis was easily supported with the data so didn't require much in manipulations. Used stacked bar graph; Reduced Scale; Reworded labels.



Step 5: Averages Table

- Given your selected protected class variable (independent variable), calculate the average (using mean, median, and mode) of the protected class group.
 - Hint: Variables might need to be converted to numerical values as needed. If you do so, please provide the mapping used.
- Run the random sampling method using 50% of the data to create a reduced dataset. Calculate the average (mean, median, and mode) of the protected class group. Indicate if there is a difference (or not) between the original dataset and the reduced dataset for any of the averages.

Report output: In your report, provide your results:

- Table of averages (mean, median, mode) for the original dataset, reduced dataset, and differences between the two.

Example Output Illustrating Possible Average Computations:

Protected Class Variable (Pregnant)	Mean	Median	Mode
Original Data Set	0 (NO)	0 (NO)	0 (NO)
Reduced Data Set	0 (NO)	1 (YES)	0 (NO)
Difference	No Difference	Difference	No Difference

Step 6: Reduced Dataset

Given the reduced dataset from Step 5, **repeat Step 3** to create a frequency table and histogram using your selected dependent variable as a function of your selected independent variable (*from Step 4*). Explain any differences in at least 2 sentences. If you used a random sampling method, would members associated with the protected class variable benefit or be harmed? Explain your reasoning in at least 2 sentences.

Report output: In your report, provide your results:

- Frequency table and histogram for reduced dataset
- Explanation of differences (if any)
- Harmed members of protected class (if any) + reasoning

Step 7: Submission

Submit a PDF report documenting your outputs for each question.

- The report must follow JDF formatting.**

- You can find a link to the JDF template here: [JDF Templates](#). The Microsoft Word template is recommended to ensure proper formatting and styling.
 - Reports that do not follow JDF formatting, or that are not neat and well organized, may receive up to a 10% deduction, regardless of content quality.
- **Name your file exactly as follows: GTuserName_Assignment_2.pdf**
 - *Example: gBurdell3_Assignment_2.pdf*
 - Deductions may be applied if the file name is incorrect or does not follow this convention.
- **All charts, graphs, and tables must be generated using Python, Excel, or another suitable non-AI software application.**
 - Points may be deducted for charts or tables generated using AI tools.
- If you used a Jupyter Notebook (.ipynb) for your work, you may submit it as a supplementary file. However, note that **PDF report is required** and will be used for grading.